

Customer Segmentation Using K-Means Clustering on Credit Card Data

Grouping bank customers by behavior to tailor offers and reduce churn

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Why segment credit card customers?

Churn

A customer stops using their credit card.

Problem

Banks lose customers when offers and engagement are poorly targeted.

Goal

Group customers by behavior, then tailor offers to each group to reduce churn risk.



Dataset overview

~10,000

bank credit card customers (bank churn dataset)

Demographics

Age, gender, education, marital status, income category

Account and behavior

Credit limit, revolving balance, transaction amount and count, utilization ratio, contacts with the bank

Churn label

Attrition_Flag: 0 = attrited, 1 = existing. Kept out of clustering and used afterward to read each segment.

Preprocessing

1

Remove

Client ID and the pre-computed classifier columns

2

Encode

Categorical columns such as gender and education

3

Scale

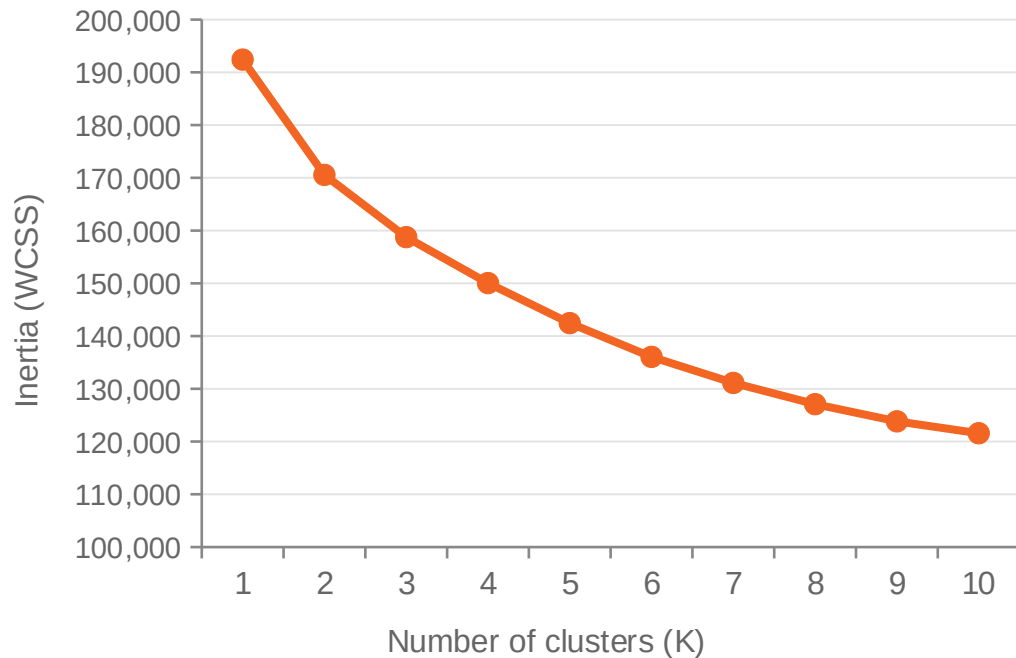
Standardize numeric features so no single unit dominates

```
df_cleaned = df.drop(columns=[
    "CLIENTNUM", <classifier cols>
])

for col in categorical_cols:
    le = LabelEncoder()
    df_cleaned[col] = le.fit_transform(
        df_cleaned[col])

features = df_cleaned.drop(
    columns=["Attrition_Flag"])
features_scaled = StandardScaler()
    .fit_transform(features)
```

Choosing K with the elbow method



K = 5

inertia 142,460

K-Means was run for K = 1 to 10. Inertia keeps falling without a sharp bend, so I chose K = 5: the point where the gains start to shrink and the groups are still distinct enough to act on.

Cluster profiles

Cluster	Avg credit limit	Avg transaction amount	Avg transaction count	Avg utilization	Retained
0	\$26,343	\$6,610	75	6%	84%
1	\$4,745	\$3,415	60	7%	67%
2	\$13,477	\$2,967	55	11%	83%
3	\$12,309	\$12,431	103	19%	96%
4	\$3,247	\$3,644	64	59%	92%

Orange marks the standout value in a column. Retained = share of customers still active (Attrition_Flag averaged by cluster).

Five customer segments

0

**High limit,
light use**

Largest limits (\$26K) on premium-tier cards, but only 6% utilization.

1

**Basic card,
highest churn**

Only 67% retained, the weakest of any group. Low limits (~\$4.7K).

2

Light spenders

Fewest transactions (55) and lowest spend (\$3.0K) despite a \$13K limit.

3

Power users

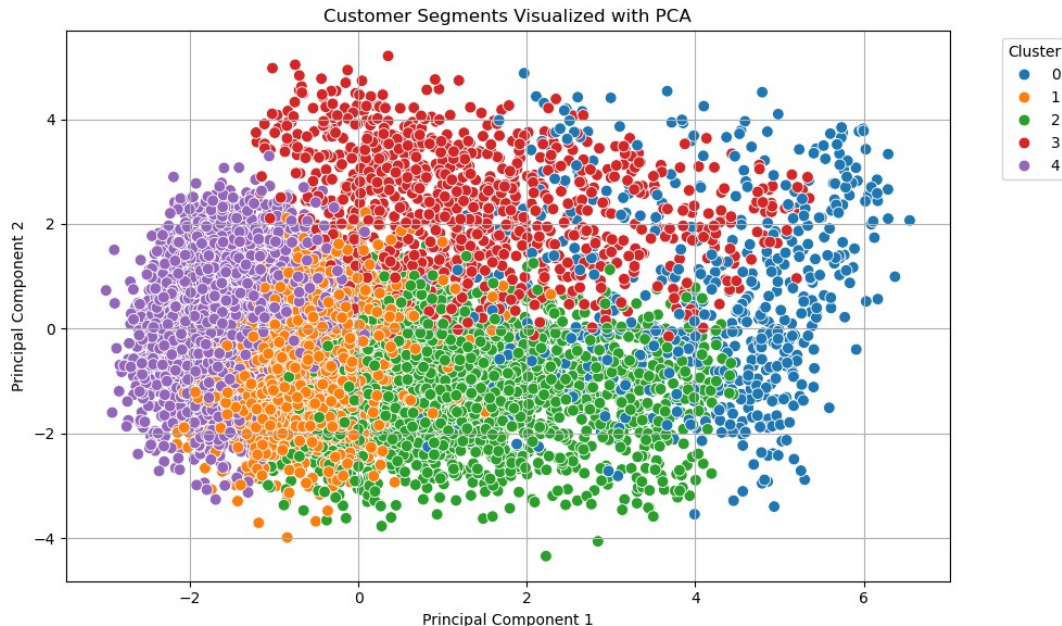
Highest spend (\$12.4K) and activity (103 transactions), 96% retained.

4

**Low limit, high
utilization**

Smallest limits (\$3.2K) with 59% of credit in use.

Seeing the clusters with PCA



Each point is a customer, projected onto the first two principal components and colored by cluster.

The groups form clear regions with some overlap at the edges, which fits segments that blend into each other rather than separating cleanly.

Recommendations by segment

Cluster 0

Reward upgrades and usage incentives

Big limits sit idle.
Give a reason to spend.

Cluster 1

Retention outreach and upgrade offers

Highest attrition.
Reach out before they leave.

Cluster 2

Cashback to lift activity

Lowest spend and transaction count.

Cluster 3

Priority retention and premium upsell

Highest value.
Protect the relationship.

Cluster 4

Budget-friendly and credit-building options

High utilization on small limits.

Limitations and next steps

Encode categories properly

Label encoding gives income and education an arbitrary numeric order. One-hot encoding would keep distances meaningful.

Check K a second way

Add silhouette scores next to the elbow plot to confirm five clusters.

Report cluster sizes

Show how many customers sit in each segment, so offers can be sized.

Test the recommendations

Run a small retention campaign per segment and compare attrition against a control group.